

Pre-lab: 8.

Design the three current mirror OTA of Figure 8-4 to obtain the following specifications:

$$G_m > 700 \mu A/V$$

$$\text{Slew Rate} > 8 V/\mu s$$

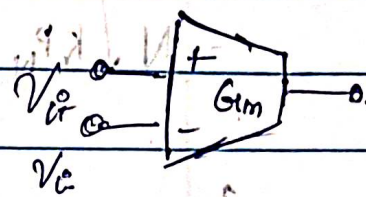
$$\text{Peak to Peak Output Swing} > 1.2 V$$

$$\text{Load Capacitance} = 20 pF$$

$$\text{Power} < 500 \mu W$$

$$\text{Power Supply} \rightarrow V_{DD} = -V_{SS} = 0.9 V$$

Symbol,



$$k_{PN} = 214.3 \frac{\mu A}{V^2}$$

$$SR > 8 V/\mu s$$

$$\text{Let } N=2$$

$$k_{DP} = 47.52$$

$$\frac{N |I_{tail}|}{C_L} > 8 V/\mu s$$

$$C_L$$

$$|I_{tail}| > \frac{8 V}{\mu s} \times 20 pF \times \frac{1}{N}$$

$$|I_{tail}| > \frac{1.6 \times 10^{-4}}{N} \times \text{lower limit of } |I_{tail}|$$

$$\text{Power} = V \cdot I \Rightarrow I = I_{DD} = V_{DD} - V_{SS} = 0.9 + 0.9 = 1.8$$

$$= (1.8 \times I) < 500 \mu W$$

$$(N+1) \frac{I}{N} < \frac{500 \mu W}{1.8}$$

$$I < \frac{277.778 \mu A}{(N+1)}$$

$$\text{Try } N=2$$

$$8 \times 10^{-5} < I < 9.259 \times 10^{-5}$$

$$80 \mu A < I < 92.59 \mu A$$

Let I_{tail} be $90 \mu A$

of H_2 output to H_2 output current source I_{tail}

$$G_m = N g_m(1,2) =$$

from lab 1

$$= N \sqrt{k_p N \left(\frac{CO}{L}\right)_{1,2} I_{tail}}$$

$$k_{pN} = 214.3$$

$$k_{pP} = 47.5$$

$$W = 5 \mu m$$

$$g_m(1,2) = \sqrt{(214.3) \left(\frac{500 nm}{270 nm}\right) 90 \mu A}$$

$$L = 270 nm$$

$$g_m(1,2) = \sqrt{5.14 \times 10^{-4}}$$

$$g_m(1,2) = 0.0188988 \times 10^{-3}$$

$$= 1.88 \times 10^{-4}$$

$$G_m = N g_m(1,2)$$

$$= 1.88 \times 10^{-4}$$

$$G_m = 0.00376$$

8.1 = We need all transistors in saturation for ~~maximum~~ maximum swing

$$V_o \text{ connected} \rightarrow M_8, M_6 \left(\frac{I}{2} \times 8.1 \right) =$$

$$8.1 > P$$

$$311.112 > P$$

$$2.01 \times 10^3 > P > 2.01 \times 8$$

for M8.

$$V_{DD} - V_o > V_{ov}(M6)$$

and $V_{DD} - V_{SS} > V_{ov}(M8)$

$$V_o < V_{DD} - V_{ov}(M6)$$

$$V_o > V_{ov}(M8) = V_{SS}$$

DC operating point

$$V_{ov}(M8) + V_{SS} < V_o < V_{DD} - V_{ov}(M6)$$

$$V_{Range} = V_{DD} - V_{ov}(M6) - (V_{ov}(M8) + V_{SS})$$

$$= 1.8 - V_{ov}(M6) - V_{ov}(M8)$$

assume

$$V_{ov} = 0.13 \text{ V}$$

$$\left(\frac{W}{L}\right)_1 = \left(\frac{W}{L}\right)_2 = \left(\frac{5 \mu\text{m}}{0.27}\right)$$

when $V_p = 0$,

$$N = 2 = \frac{(W/L)_{1,2}}{(W/L)_{3,4}}$$

$$V_{ov}(1,2) = V_{GS}(1,2) - V_{tn}$$

$$V_{GS}(1,2) = V_{ov} + V_{tn} = 0.13 + 0.4$$

$$V_{GS} = 0.53 \text{ V}$$

$$V_{GS} = V_G - V_S \Rightarrow V_G = V_S = 0$$

$$V_S = -0.53 \text{ V}$$

$$V_{DS(10)} = V_D - V_S$$

$$= V_S(1,2) - V_S(10) = -0.53 \text{ V} + 0.9 = 0.37 \text{ V}$$

$$V_{DS} < V_{DS(10)} \quad 0.37V$$

We want $V_{DS} > V_{ov}$

$$0.37 > 0.13V \quad \checkmark \text{ satisfied}$$

$$I_D = \frac{1}{2} \mu_n C_{ox} \left(\frac{W}{L} \right)_{10} (V_{ov10})^2$$

$$I_D = 90 \mu A$$

$$90 \mu A = \frac{1}{2} (214.3) \left(\frac{W}{L} \right)_{10} (0.13)^2$$

$$\left(\frac{W}{L} \right)_{10} = 21.53$$

$$I_{D2} = \frac{I_{tail}}{2} = 45 \mu A$$

M_4, M_6 are current Mirrors with ratio 2.

$$I_{D6} = I_{D(8)} = I_{tail}$$

$$I_{D8} = I_{tail} = \frac{1}{2} \mu_n C_{ox} \left(\frac{W}{L} \right)_8 (V_{ov})^2$$

$$0.09V = 0.13V$$

$$V_{DS} = 0.13V$$

$$2V = 0.13V$$

$$\left(\frac{W}{L}\right)_8 = \frac{90\mu \times 2}{\mu_n C_{ox} (0.13)^2}$$

$$\boxed{\left(\frac{W}{L}\right)_8 = 49.7 \approx 50.}$$

$$\left(\frac{W}{L}\right)_{1,2} = \frac{5\mu m}{0.27\mu m}$$

$$\left(\frac{W}{L}\right)_{3,4} = \frac{5\mu m}{0.27\mu m}$$

1:2 Current Mirror.

$$\left(\frac{W}{L}\right)_5 = \left(\frac{W}{L}\right)_6 = 2 \times \frac{5}{0.27\mu m} = \frac{10\mu m}{0.27\mu m}$$

$$\left(\frac{W}{L}\right)_7 = \left(\frac{W}{L}\right)_8 = 50 = \frac{27 \times 5\mu m}{0.27\mu m}$$

$$\left(\frac{W}{L}\right)_7 = \left(\frac{W}{L}\right)_8 = \left(\frac{W}{L}\right)_9 = \left(\frac{W}{L}\right)_{10} = 22 \Rightarrow 11.88 \times 5 / 0.27 \approx (12 \times 5 / 0.27\mu m)$$

~~22~~